

AI Unit Calculator: Render Deployment Instructions

Use this checklist to deploy the AI Unit Calculator as a public Render Web Service. This is the recommended first production host because the project already uses a Vite React frontend and an Express backend in `server.mjs`.

Before You Start

- Create or confirm a GitHub repository for the project. Render deploys from a linked GitHub, GitLab, Bitbucket, or public Git repository.
- Make sure `.env` is not committed. Production secrets should be entered in Render, not stored in GitHub.
- Use live PayPal credentials only when you are ready for real transactions. Keep sandbox credentials for local testing.

Create The Render Web Service

- Open <https://dashboard.render.com/> and sign in.
- Click New, then choose Web Service.
- Choose Git Provider and connect GitHub if prompted.
- Select the AI Unit Calculator repository.
- Complete the service form with the settings below.

```
Name: ai-unit-calculator
Language: Node
Branch: main
Build Command: npm install && npm run build
Start Command: npm run preview
```

- Choose a region near your expected users.
- Choose an instance type. Start small and upgrade later if traffic requires it.
- Open Advanced and add the environment variables listed below.

Production Environment Variables

```
PAYPAL_MODE=live
PAYPAL_CLIENT_ID=your_live_paypal_client_id
PAYPAL_CLIENT_SECRET=your_live_paypal_secret
ENABLE_DEV MOCK_PAYMENTS=false
ADMIN_PASSWORD=your_admin_password
AUTH_SECRET=a_long_random_secret_value
API_PORT=10000
WEB_PORT=10000
```

If you are using the GCP pricing refresh feature, also add the GCP billing API key and any other pricing source overrides you need.

GCP_BILLING_API_KEY=your_key_if_using_gcp_refresh

Deploy And Test

- Click Create Web Service. Render will build and deploy the app.
- When deploy completes, Render will provide a temporary URL similar to <https://ai-unit-calculator.onrender.com>.
- Open that URL and verify the home page loads.
- Go to /admin and verify the admin password works.
- Check the PayPal configuration panel and confirm live mode, server verification configured, and dev mock payments disabled.
- Perform a careful low-dollar live transaction only when you are ready to confirm real payments.

Custom Domain

After the Render URL works, add your custom domain aiunitcalculator.ai in Render. Render will provide DNS records to add at your domain registrar. Wait for DNS and HTTPS provisioning to complete, then test the custom domain.

Reference

Render Web Services documentation: <https://render.com/docs/web-services>